

UNIT-3: INDEX NUMBERS

Comprehensive Theoretical Foundations, Advanced Methodologies, Mathematical Proofs, and Extensively Solved Numerical Problems

1. Introduction to Index Numbers: Concepts and Applications

| 1.1 Theoretical Framework and Definition

An index number is a specialized statistical measure designed to facilitate comparison and track the relative changes in the magnitude of a variable or a group of related variables over time, across geographical locations, or between other distinct categories. Often referred to as "economic barometers," index numbers are indispensable analytical devices in empirical economics, business management, policy analysis, and financial markets. They distill multi-dimensional, complex real-world data into a singular, readily interpretable metric that reflects aggregate fluctuations.

Formally, an index number expresses the value of a given variable in the current period as a percentage relative to its value in a designated baseline period. It is crucial to underscore that index numbers are specialized averages. While standard descriptive statistical tools like the arithmetic mean or median summarize observations with uniform dimensions, index numbers aggregate composite items measured across disparate physical units (e.g., kilograms of wheat, liters of milk, meters of textile, and kilowatt-hours of electrical energy). By converting absolute quantities and monetary expressions into dimensionless relative values, they allow analytical comparisons to be drawn uniformly.

| 1.2 Structural Characteristics of Index Numbers

To fully grasp the operational capability of index numbers, one must examine their distinct structural attributes:

1. **Specialized Averages:** Standard measures of central tendency can only be computed when data are completely homogeneous and expressed in identical units. In contrast, index numbers function as a specialized framework capable of averaging movements across highly heterogeneous groups of commodities.
2. **Measurement of Relative Change:** Rather than documenting absolute variation, index numbers measure the percentage or proportional changes between two chronological periods or geographical environments. For instance, knowing that the absolute cost of living rose by \$500 provides minimal analytical insight without establishing the initial baseline expenditure.
3. **Dimensionless and Pure Numbers:** Because they are constructed as ratios multiplied by a scalar (typically 100), the final figures are independent of the underlying physical units of measurement.
4. **Aggregation of Complex Phenomena:** Index numbers quantify macro-level trends that cannot be observed or measured directly, such as inflation, industrial productivity, purchasing power, and standard of living.

1.3 Historical Evolution and Economic Utility

The origin of index numbers dates back to the 18th century, with early conceptual frameworks developed by Italian economist Carli in 1764 to analyze the impact of the discovery of America on European prices. Over the subsequent centuries, the mathematical architecture of index numbers was refined by seminal statisticians including Laspeyres, Paasche, Fisher, Marshall, Edgeworth, and Bowley.

In contemporary macroeconomic administration, index numbers serve several critical functions:

- **Policy Formulation:** Central banks and ministries of finance rely on Consumer Price Indices (CPI) and Wholesale Price Indices (WPI) to engineer monetary policies, adjust interest rates, and design fiscal programs.
- **Deflation of Nominal Values:** Index numbers are extensively deployed to convert nominal economic parameters (such as nominal GDP, nominal wages, and nominal revenue) into real, inflation-adjusted values, enabling valid long-term growth assessments.
- **Adjustment of Wages and Allowances:** Modern employment contracts and public sector compensation schemes incorporate Dearness Allowance (DA) and cost-of-living adjustments directly tied to CPI variations, preserving real purchasing power.
- **Business Forecasting:** Corporate entities utilize sector-specific index numbers (e.g., indices of industrial production, real estate price indices, and retail sales indices) to forecast market demand, allocate capital, and manage supply chain vulnerabilities.

1.4 Nomenclature and Mathematical Notation

To ensure algebraic rigor throughout these extensive notes, we establish a standardized notation system that will govern all subsequent chapters:

- P_0 = Price of an individual commodity or item in the chosen **base period**.
- P_1 = Price of an individual commodity or item in the **current period** (the period under investigation).
- Q_0 = Quantity or volume of an individual commodity produced, consumed, or distributed in the **base period**.
- Q_1 = Quantity or volume of an individual commodity produced, consumed, or distributed in the **current period**.
- W = Weight or relative importance coefficient assigned to a specific commodity within the composite basket.
- P_{01} = Price index number for the current period (**1**) relative to the base period (**0**).
- Q_{01} = Quantity index number for the current period (**1**) relative to the base period (**0**).
- V_{01} = Value index number for the current period (**1**) relative to the base period (**0**).
- Σ = The summation operator, denoting the cumulative addition across all items within the chosen market basket.

1.5 The Methodological Challenges in Index Number Construction

Constructing a mathematically sound and practically meaningful index number requires resolving several foundational, interconnected structural challenges:

1.5.1 Delineating the Core Purpose of the Index

There is no universally applicable index number. An index designed to monitor wholesale industrial raw material costs is entirely unsuited for adjusting the compensation of agricultural laborers. The structural purpose dictates every subsequent parameter, including the choice of items, base period, and mathematical formula.

1.5.2 Selection of the Commodity Basket

It is physically and logistically impossible to monitor every single item traded within an economy. Therefore, a representative sample or "basket" must be curated. The chosen items must satisfy the following criteria:

- They must reflect the consumption or production realities of the target population.
- They must be standard, clearly identifiable, and non-generic grades of commodities.
- They must possess stable demand patterns over reasonable time horizons.

1.5.3 Sourcing Reliable Price and Quantity Quotations

Price data varies dramatically depending on geographical location, transaction scale (wholesale vs. retail), and seasonal factors. To avoid systemic bias, data collection networks must secure reliable, standard price quotations from selected markets at regular intervals.

1.5.4 Choice of the Base Period

The base period serves as the structural anchor point for the index. A properly selected base period must satisfy two fundamental constraints:

1. **Economic Normality:** The base year must be economically stable, free from extreme shocks, black swan events, pandemics, war, severe hyperinflation, or natural catastrophes.
2. **Temporal Proximity:** The base period should not be too remote in the past. If a base period is decades old, changing consumer tastes, structural technological shifts, and the emergence of entirely new product classes render comparisons obsolete.

Furthermore, analysts must decide between a **Fixed Base System** (where the reference point remains constant over time) and a **Chain Base System** (where the preceding year continuously serves as the base for the current year).

1.5.5 Selection of the Mathematical Formula and Weighting Architecture

The choice of mathematical formula is critical, as different methods yield diverging results from identical source data. Statisticians must determine whether to use simple unweighted structures or sophisticated weighted formulations that reflect the relative economic importance of each item. This choice involves balancing computational simplicity against structural accuracy, as explored in the following sections.

2. Unweighted Index Numbers

| 2.1 Theoretical Overview

Unweighted index numbers operate under the simplifying assumption that all commodities within the selected basket possess equal economic importance. No explicit weights or significance multipliers are assigned to individual items. While this approach dramatically reduces data requirements and simplifies calculation, it introduces substantial structural distortions, as minor commodities exert an identical mathematical influence on the final index as critical, high-volume items (e.g., treating table salt and wheat with identical significance).

Unweighted index numbers are divided into two primary mathematical methodologies: the **Simple Aggregative Method** and the **Simple Average of Price Relatives Method**.

| 2.2 Simple Aggregative Method

In the Simple Aggregative Method, the absolute prices of all items in the current period are summed and expressed as a direct percentage of the sum of their absolute prices in the designated base period.

The mathematical formulation is expressed as follows:

$$P_{01} = (\Sigma P_1 / \Sigma P_0) \times 100$$

Where:

ΣP_1 = Total sum of the prices of all commodities in the current period.

ΣP_0 = Total sum of the prices of all commodities in the base period.

2.2.1 Critical Critique of the Simple Aggregative Method

Although highly straightforward, this method suffers from a severe structural flaw: **it is heavily dependent on the physical units of measurement**. Commodities with higher absolute prices per unit exert a disproportionate impact on the index, regardless of their actual consumption volume or economic relevance. For example, if Commodity A is priced in dollars per ounce and Commodity B in dollars per ton, changing the unit scale completely alters the final index value. Consequently, this method is rarely utilized in professional applications.

| 2.3 Simple Average of Price Relatives Method

To overcome the unit-dependency limitation of the Simple Aggregative Method, the Simple Average of Price Relatives Method converts the absolute price of each individual commodity into a dimensionless ratio, known as a **price relative**, before calculating an aggregate average. The price relative for an individual item is defined as:

$$\text{Price Relative (R)} = (P_1 / P_0) \times 100$$

Once individual price relatives are computed, they can be averaged across the basket using either the **Arithmetic Mean (AM)** or the **Geometric Mean (GM)**.

2.3.1 Computation Using the Arithmetic Mean

When employing the arithmetic mean as the averaging mechanism, the individual price relatives are summed and divided by the total number of items (N):

$$P_{01} \text{ (using AM)} = \Sigma [(P_1 / P_0) \times 100] / N = \Sigma R / N$$

2.3.2 Computation Using the Geometric Mean

The geometric mean is widely considered a superior mathematical average for ratios and percentage changes. It provides a more balanced tracking of multi-directional price movements and satisfies key mathematical consistency tests more effectively than the arithmetic mean. Its formulation is:

$$P_{01} \text{ (using GM)} = \text{Antilog} [\Sigma \log R / N]$$

Or alternatively, expressed directly through price ratios:

$$P_{01} \text{ (using GM)} = \text{Antilog} [\Sigma \log (P_1 / P_0) / N] \times 100$$

| 2.4 Comprehensive Solved Numerical Problems (Unweighted)

Problem 2.1: Simple Aggregative Method vs. Simple Average of Relatives (AM & GM)

Given the following structural retail price data for five essential commodities for the years 2020 (Base Year) and 2025 (Current Year), calculate the Price Index Numbers using:

1. Simple Aggregative Method
2. Simple Average of Price Relatives using Arithmetic Mean
3. Simple Average of Price Relatives using Geometric Mean

Commodity	Unit Measurement	Price in 2020 (P_0) (\$)	Price in 2025 (P_1) (\$)
A (Wheat)	Per Kg	20	26
B (Milk)	Per Liter	40	60
C (Sugar)	Per Kg	30	36
D (Clothing)	Per Meter	100	130
E (Fuel)	Per Liter	80	90

Solution:

Let us construct an integrated analytical computation table to derive all necessary components.

Item	P_0	P_1	$R = (P_1/P_0) \times 100$	$\log R$
A	20	26	$(26/20) \times 100 = 130.00$	2.11394
B	40	60	$(60/40) \times 100 = 150.00$	2.17609
C	30	36	$(36/30) \times 100 = 120.00$	2.07918
D	100	130	$(130/100) \times 100 = 130.00$	2.11394
E	80	90	$(90/80) \times 100 = 112.50$	2.05115
Total (Σ)	$\Sigma P_0 = 270$	$\Sigma P_1 = 342$	$\Sigma R = 642.50$	$\Sigma \log R = 10.53430$

Step 1: Simple Aggregative Method Calculation

$$P_{01} = (\Sigma P_1 / \Sigma P_0) \times 100 = (342 / 270) \times 100 = 1.26667 \times 100 = 126.67$$

Interpretation: According to the simple aggregative method, the aggregate price level of this commodity group increased by 26.67% over the five-year period.

Step 2: Simple Average of Price Relatives using Arithmetic Mean

Number of items (N) = 5.

$$P_{01} \text{ (AM)} = \Sigma R / N = 642.50 / 5 = 128.50$$

Interpretation: The simple arithmetic average of individual price changes yields a total index level of 128.50, representing an average price appreciation of 28.50%.

Step 3: Simple Average of Price Relatives using Geometric Mean

$$P_{01} \text{ (GM)} = \text{Antilog} [\Sigma \log R / N] = \text{Antilog} [10.53430 / 5] = \text{Antilog} [2.10686]$$

Calculating the antilog of 2.10686:

$$P_{01} \text{ (GM)} = 127.89$$

Interpretation: The geometric mean provides a slightly lower, more accurate tracking of relative adjustments at 127.89, corresponding to a 27.89% increase.

Problem 2.2: Demonstrating Unit Dependency Distortion

To demonstrate why the Simple Aggregative Method is mathematically unreliable, consider two commodities, X and Y. We will compute the index under Scenario I (Standard units) and Scenario II (where the physical measurement scale of X is altered), while the underlying prices remain unchanged.

Scenario I: Item X is priced per Kg (\$10 in Year 0, \$20 in Year 1). Item Y is priced per Gram (\$5 in Year 0, \$5 in Year 1).

Scenario II: Item X is repackaged and priced per Metric Ton (\$10,000 in Year 0, \$20,000 in Year 1). Item Y remains unchanged.

Solution:

Calculation under Scenario I:

Item	P ₀	P ₁
X (per kg)	10	20
Y (per gram)	5	5
Total	Σ P ₀ = 15	Σ P ₁ = 25

$$P_{01} \text{ (Scenario I)} = (25 / 15) \times 100 = 166.67$$

Calculation under Scenario II:

Item	P ₀	P ₁
X (per ton)	10,000	20,000
Y (per gram)	5	5
Total	Σ P ₀ = 10,005	Σ P ₁ = 20,005

$$P_{01} \text{ (Scenario II)} = (20,005 / 10,005) \times 100 = 199.95$$

Analytical Insight: The price movement of Item Y is identical in both scenarios, yet changing the unit scale of Item X drives the index from 166.67 to 199.95. This clearly illustrates why the Simple Aggregative Method is structurally flawed for mixed-unit commodity groups.

3. Weighted Index Numbers

3.1 Theoretical Necessity for Weighting

In actual economic contexts, different goods and services occupy vastly distinct positions in consumer budgets and manufacturing supply chains. A 50% price spike in petroleum or basic food staples has immediate, systemic macroeconomic consequences, whereas an identical 50% increase in the price of writing ink or fine chinaware is relatively marginal. Unweighted calculations fail to capture this distinction. To construct an accurate representation of price movements, each component must be assigned a mathematical weight proportional to its economic importance.

Weights are typically assigned based on either the monetary value of consumption or production volumes during a specified period. Weighted index numbers are broadly classified into two categories: **Weighted Aggregative Index Numbers** and **Weighted Average of Price Relatives**.

3.2 Weighted Aggregative Index Numbers

This approach weights absolute prices using quantity or value parameters from either the base period, the current period, or an average of both. We will analyze the core formulations below.

3.2.1 Laspeyres' Price Index (Base-Year Weighting Architecture)

Developed by German economist Etienne Laspeyres in 1871, this formulation utilizes the quantities consumed during the **base period** (Q_0) as the fixed weighting factor. It measures the changing cost of purchasing a fixed base-year basket of goods over time.

$$P_{01}^L = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100$$

Operational Advantage: Because the weights are derived entirely from base-year data, quantities only need to be surveyed once. This makes the index highly cost-effective and straightforward to update on a continuous basis.

Structural Limitation: It does not account for changes in consumption patterns over time. Typically, consumers substitute cheaper alternatives for items experiencing rapid price increases. By holding base-year quantities constant, the Laspeyres index tends to overstate the true cost of living increase (often referred to as the **Laspeyres upward bias**).

3.2.2 Paasche's Price Index (Current-Year Weighting Architecture)

Developed by Hermann Paasche in 1874, this approach utilizes the quantities consumed during the **current period** (Q_1) as the weighting factor. It evaluates the current-period consumption basket under both current and base-period prices.

$$P_{01}^P = (\Sigma P_1 Q_1 / \Sigma P_0 Q_1) \times 100$$

Operational Advantage: It reflects contemporary consumption trends and structural shifts in consumer preferences more accurately than base-year weighting systems.

Structural Limitation: Gathering complete, high-fidelity quantity and volume data for every current tracking period is logistically demanding and expensive. Furthermore, by embedding current substitution behaviors, the Paasche index tends to understate cost increases (referred to as the **Paasche downward bias**).

3.2.3 Marshall-Edgeworth Index

This approach attempts to balance the structural frameworks of Laspeyres and Paasche by utilizing the arithmetic mean of the base-period and current-period quantities as the weight:

$$\text{Weight (Q)} = (Q_0 + Q_1) / 2$$

Substituting this weight into the aggregative equation removes the divisor, yielding the following formulation:

$$P_{01}^{ME} = [\Sigma P_1 (Q_0 + Q_1) / \Sigma P_0 (Q_0 + Q_1)] \times 100 = [(\Sigma P_1 Q_0 + \Sigma P_1 Q_1) / (\Sigma P_0 Q_0 + \Sigma P_0 Q_1)] \times 100$$

3.2.4 Bowley's Index (Drobisch-Bowley Formulation)

Bowley's index takes the direct arithmetic mean of the Laspeyres and Paasche index numbers, attempting to balance their respective upward and downward biases:

$$P_{01}^B = (P_{01}^L + P_{01}^P) / 2$$

3.3 Weighted Average of Price Relatives Index

In this framework, individual price relatives (**R**) are computed for each commodity and then aggregated using value-based weights (**V = P × Q**). This can be structured using either base-year values or current-year values.

Using base-year value weights (**W = P₀Q₀**) with an arithmetic mean framework:

$$P_{01} = \Sigma R W / \Sigma W$$

Where **R = (P₁ / P₀) × 100** and **W = P₀Q₀**. Substituting these terms demonstrates that this method is algebraically identical to the Laspeyres index:

$$\Sigma R W = \Sigma [(P_1 / P_0) \times 100 \times P_0 Q_0] = 100 \times \Sigma P_1 Q_0$$

$$\Sigma R W / \Sigma W = (100 \times \Sigma P_1 Q_0) / \Sigma P_0 Q_0 = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100$$

| 3.4 Comprehensive Solved Numerical Problems (Weighted)

Problem 3.1: Complete Evaluation of Weighted Aggregative Forms

Using the data table below, calculate the following index numbers for the current year:

1. Laspeyres' Price Index
2. Paasche's Price Index
3. Marshall-Edgeworth Price Index
4. Bowley's Price Index

Item	Base Year (0)		Current Year (1)	
	Price (P_0)	Quantity (Q_0)	Price (P_1)	Quantity (Q_1)
X	10	20	15	22
Y	12	15	14	18
Z	8	30	12	32
W	5	40	6	38

Solution:

We construct an analytical calculation table to compute the four primary cross-product summation terms:

P_1Q_0 , P_0Q_0 , P_1Q_1 , and P_0Q_1 .

Item	P_0	Q_0	P_1	Q_1	P_1Q_0	P_0Q_0	P_1Q_1	P_0Q_1
X	10	20	15	22	$15 \times 20 = 300$	$10 \times 20 = 200$	$15 \times 22 = 330$	$10 \times 22 = 220$
Y	12	15	14	18	$14 \times 15 = 210$	$12 \times 15 = 180$	$14 \times 18 = 252$	$12 \times 18 = 216$
Z	8	30	12	32	$12 \times 30 = 360$	$8 \times 30 = 240$	$12 \times 32 = 384$	$8 \times 32 = 256$
W	5	40	6	38	$6 \times 40 = 240$	$5 \times 40 = 200$	$6 \times 38 = 228$	$5 \times 38 = 190$
Total (Σ)					810	440	776	392

Step 1: Laspeyres' Index Calculation

$$P_{01}^L = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100 = (1110 / 820) \times 100$$

Recalculating with the correct values from our table totals:

$$P_{01}^L = (1110 / 820) [\text{Correction: } \Sigma P_1 Q_0 = 1110, \Sigma P_0 Q_0 = 820]$$

Let us re-verify the table sums directly from our computed data:

$$\Sigma P_1 Q_0 = 300 + 210 + 360 + 240 = 1110$$

$$\Sigma P_0 Q_0 = 200 + 180 + 240 + 200 = 820$$

$$\Sigma P_1 Q_1 = 330 + 252 + 384 + 228 = 1194$$

$$\Sigma P_0 Q_1 = 220 + 216 + 256 + 190 = 882$$

$$P_{01}^L = (1110 / 820) \times 100 = 135.37$$

Step 2: Paasche's Index Calculation

$$P_{01}^P = (\Sigma P_1 Q_1 / \Sigma P_0 Q_1) \times 100 = (1194 / 882) \times 100 = 135.37$$

Step 3: Marshall-Edgeworth Index Calculation

$$P_{01}^{ME} = [(\Sigma P_1 Q_0 + \Sigma P_1 Q_1) / (\Sigma P_0 Q_0 + \Sigma P_0 Q_1)] \times 100$$

$$P_{01}^{ME} = [(1110 + 1194) / (820 + 882)] \times 100 = [2304 / 1702] \times 100 = 135.37$$

Step 4: Bowley's Index Calculation

$$P_{01}^B = (P_{01}^L + P_{01}^P) / 2 = (135.37 + 135.37) / 2 = 135.37$$

Observation: Due to proportional values in this specific dataset, the derived index numbers converge uniformly to 135.37, indicating a consistent 35.37% rise across all weighting variations.

4. Fisher's Ideal Index Number

4.1 Conceptualization and Definition

Developed by American mathematical economist Irving Fisher, **Fisher's Ideal Index Number** is defined as the **geometric mean of the Laspeyres and Paasche index numbers**. Fisher designated this formulation as "ideal" because it satisfies rigorous mathematical and economic criteria that few other formulations meet.

Its algebraic expression is:

$$P_{01}^F = \sqrt{(P_{01}^L \times P_{01}^P)}$$

Expanding this into its fundamental components yields:

$$P_{01}^F = \sqrt{[(\sum P_1 Q_0 / \sum P_0 Q_0) \times (\sum P_1 Q_1 / \sum P_0 Q_1)]} \times 100$$

4.2 Attributes and Advantages of Fisher's Formulation

Fisher's index stands out in index number theory due to several structural features:

1. **Geometric Mean Integration:** By utilizing the geometric mean to combine the Laspeyres and Paasche formulas, it balances their respective upward and downward biases.
2. **Dual-Period Quantity Attribution:** It incorporates quantity data from both the base and current periods, capturing structural shifts in consumption patterns.
3. **Reversibility:** It satisfies both the Time Reversal Test and the Factor Reversal Test, which serve as core benchmarks for mathematical consistency in index construction.

4.3 Mathematical Axioms: Consistency Tests

To evaluate the mathematical rigor of an index formula, statisticians utilize three standard consistency tests: the Time Reversal Test, the Factor Reversal Test, and the Circular Test.

4.3.1 The Time Reversal Test (TRT)

The Time Reversal Test requires that if the base period and current period are chronologically interchanged, the resulting index should be the reciprocal of the original index. Formally:

$$P_{01} \times P_{10} = 1$$

Where P_{10} represents the index for the base period computed relative to the current period as base.

Proof for Fisher's Ideal Index:

We know that:

$$P_{01} = \sqrt{[(\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times (\Sigma P_1 Q_1 / \Sigma P_0 Q_1)]}$$

Interchanging the time subscripts 0 and 1 gives:

$$P_{10} = \sqrt{[(\Sigma P_0 Q_1 / \Sigma P_1 Q_1) \times (\Sigma P_0 Q_0 / \Sigma P_1 Q_0)]}$$

Multiplying the two expressions:

$$P_{01} \times P_{10} = \sqrt{[(\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times (\Sigma P_1 Q_1 / \Sigma P_0 Q_1) \times (\Sigma P_0 Q_1 / \Sigma P_1 Q_1) \times (\Sigma P_0 Q_0 / \Sigma P_1 Q_0)]}$$

All internal ratios cancel out under the radical:

$$P_{01} \times P_{10} = \sqrt{1} = 1$$

Thus, Fisher's index satisfies the Time Reversal Test.

4.3.2 The Factor Reversal Test (FRT)

The Factor Reversal Test requires that interchanging the price and quantity variables within an index formula should yield a quantity index, and that the product of the price and quantity indices should equal the true ratio of total values. Formally:

$$P_{01} \times Q_{01} = \Sigma P_1 Q_1 / \Sigma P_0 Q_0$$

Proof for Fisher's Ideal Index:

The quantity counterpart of Fisher's price index (Q_{01}) is constructed by swapping P and Q :

$$Q_{01} = \sqrt{[(\Sigma Q_1 P_0 / \Sigma Q_0 P_0) \times (\Sigma Q_1 P_1 / \Sigma Q_0 P_1)]}$$

Multiplying P_{01} and Q_{01} :

$$P_{01} \times Q_{01} = \sqrt{[(\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times (\Sigma P_1 Q_1 / \Sigma P_0 Q_1) \times (\Sigma P_0 Q_1 / \Sigma P_0 Q_0) \times (\Sigma P_1 Q_1 / \Sigma P_1 Q_0)]}$$

Simplifying the expression by canceling identical terms in the numerator and denominator:

$$P_{01} \times Q_{01} = \sqrt{[(\Sigma P_1 Q_1)^2 / (\Sigma P_0 Q_0)^2]} = \Sigma P_1 Q_1 / \Sigma P_0 Q_0$$

Thus, Fisher's index satisfies the Factor Reversal Test. (Notably, neither Laspeyres nor Paasche passes this test).

4.3.3 The Circular Test

The Circular Test is an extension of the Time Reversal Test for scenarios involving more than two periods. It requires that an index should remain consistent when tracking shifts across a sequence of periods (e.g., from period 0 to 1, 1 to 2, and back to 0). Formally:

$$P_{01} \times P_{12} \times P_{20} = 1$$

Fisher's Ideal Index **does not satisfy** the Circular Test. The primary formulations that pass this test are unweighted geometric means of price relatives and fixed-weight aggregative indices (such as Kelly's index).

| 4.4 Comprehensive Solved Numerical Problems (Fisher)

Problem 4.1: Computing Fisher's Ideal Index and Verifying TRT & FRT

Using the data table below, compute Fisher's Ideal Price Index and formally verify that it satisfies both the Time Reversal Test and the Factor Reversal Test.

Commodity	Base Year		Current Year	
	Price (P_0)	Quantity (Q_0)	Price (P_1)	Quantity (Q_1)
A	4	10	5	12
B	3	15	4	20
C	6	8	8	10
D	5	12	6	14

Solution:

We construct the cross-product table:

Item	P_0	Q_0	P_1	Q_1	P_1Q_0	P_0Q_0	P_1Q_1	P_0Q_1
A	4	10	5	12	50	40	60	48
B	3	15	4	20	60	45	80	60
C	6	8	8	10	64	48	80	60
D	5	12	6	14	72	60	84	70
Σ Totals					246	193	304	238

Step 1: Compute Fisher's Ideal Index

$$P_{01}^F = \sqrt{[(246 / 193) \times (304 / 238)]} \times 100$$

$$P_{01}^F = \sqrt{[1.27461 \times 1.27731]} \times 100 = \sqrt{1.62807} \times 100 = 1.27596 \times 100 = 127.60$$

Step 2: Verification of the Time Reversal Test

$$P_{10} = \sqrt{[(\Sigma P_0Q_1 / \Sigma P_1Q_1) \times (\Sigma P_0Q_0 / \Sigma P_1Q_0)]} = \sqrt{[(238 / 304) \times (193 / 246)]}$$

Multiplying P_{01} (without the 100 multiplier) and P_{10} :

$$P_{01} \times P_{10} = \sqrt{[(246 / 193) \times (304 / 238) \times (238 / 304) \times (193 / 246)]} = \sqrt{1} = 1$$

Result: The Time Reversal Test is verified successfully.

Step 3: Verification of the Factor Reversal Test

$$Q_{01} = \sqrt{[(\Sigma Q_1 P_0 / \Sigma Q_0 P_0) \times (\Sigma Q_1 P_1 / \Sigma Q_0 P_1)]} = \sqrt{[(238 / 193) \times (304 / 246)]}$$

Multiplying P_{01} and Q_{01} :

$$P_{01} \times Q_{01} = \sqrt{[(246 / 193) \times (304 / 238) \times (238 / 193) \times (304 / 246)]}$$

$$P_{01} \times Q_{01} = \sqrt{[(304 \times 304) / (193 \times 193)]} = 304 / 193$$

The true value ratio is:

$$\Sigma P_1 Q_1 / \Sigma P_0 Q_0 = 304 / 193$$

Result: The product matches the value ratio exactly. The Factor Reversal Test is verified successfully.

5. Consumer Price Index (CPI) Numbers

5.1 Concept and Significance

The **Consumer Price Index (CPI)**, also known as the **Cost of Living Index Number**, is designed to track changes over time in the prices of a fixed basket of consumer goods and services purchased by a specific target population group (e.g., urban laborers, industrial workers, or agricultural employees).

Unlike wholesale indices, the CPI evaluates the actual retail prices paid by consumers, making it a critical tool for adjusting wages, assessing currency purchasing power, and formulating national social security policies.

5.2 Methodological Approaches for CPI Construction

There are two primary mathematical frameworks used to calculate the Consumer Price Index: the **Aggregate Expenditure Method** and the **Family Budget Method**.

5.2.1 Aggregate Expenditure Method

This method mirrors the Laspeyres aggregative price index framework. It evaluates the total cost of purchasing the base-year consumption basket at current prices relative to base-year prices.

$$\text{CPI} = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100$$

5.2.2 Family Budget Method (Weighted Average of Price Relatives)

This method calculates a weighted average of individual price relatives using the total expenditure on each item in the base year as the weight. The formula is expressed as:

$$\text{CPI} = \Sigma R W / \Sigma W$$

Where:

$R = (P_1 / P_0) \times 100$ (The price relative for each commodity)

$W = P_0 Q_0$ (The base-year monetary value expenditure on that commodity)

| 5.3 Comprehensive Solved Numerical Problems (CPI)

Problem 5.1: Comparative Calculation of CPI

Given the following consumption data for an urban working-class community, calculate the Consumer Price Index using both the Aggregate Expenditure Method and the Family Budget Method to demonstrate their algebraic equivalence.

Commodity	Base Quantity (Q_0)	Base Price (P_0) (\$)	Current Price (P_1) (\$)
Food	50	10	14
Rent	1	100	150
Clothing	10	20	25
Fuel & Light	15	5	8
Miscellaneous	20	15	18

Solution:

We construct an integrated computation table containing components for both methodologies.

Item	Q_0	P_0	P_1	$P_1 Q_0$	$P_0 Q_0$ (W)	$R = (P_1/P_0) \times 100$	RW
Food	50	10	14	700	500	140.0	70,000
Rent	1	100	150	150	100	150.0	15,000
Clothing	10	20	25	250	200	125.0	25,000
Fuel	15	5	8	120	75	160.0	12,000
Misc	20	15	18	360	300	120.0	36,000
Sum (Σ)				1580	1175	-	158,000

Method 1: Aggregate Expenditure Method

$$CPI = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100 = (1580 / 1175) \times 100 = 134.47$$

Method 2: Family Budget Method

$$CPI = \Sigma R W / \Sigma W = 158,000 / 1175 = 134.47$$

Conclusion: Both analytical frameworks yield an identical index value of 134.47, illustrating that the family budget approach is an algebraic rearrangement of the aggregate expenditure framework.

6. Miscellaneous Practice Problems and Solutions

This chapter provides a wide range of fully solved numerical problems covering various applications and special cases of index numbers, designed to reinforce analytical proficiency.

Problem 6.1: Missing Frequency / Value Derivation

An incomplete weighted price index calculated using base-year weights indicates a final value of 125. Determine the missing base-year quantity (x) for Item C based on the table below.

Item	P_0	P_1	Q_0
A	8	12	10
B	10	15	5
C	5	5	x

Solution:

The index uses base-year weights, corresponding to Laspeyres' formula:

$$\text{Index} = (\Sigma P_1 Q_0 / \Sigma P_0 Q_0) \times 100 = 125$$

Let us express the sums in terms of x :

$$\Sigma P_1 Q_0 = (12 \times 10) + (15 \times 5) + (5 \times x) = 120 + 75 + 5x = 195 + 5x$$

$$\Sigma P_0 Q_0 = (8 \times 10) + (10 \times 5) + (5 \times x) = 80 + 50 + 5x = 130 + 5x$$

Set up the equation:

$$[(195 + 5x) / (130 + 5x)] = 1.25$$

$$195 + 5x = 1.25 \times (130 + 5x)$$

$$195 + 5x = 162.5 + 6.25x$$

$$195 - 162.5 = 6.25x - 5x$$

$$32.5 = 1.25x \Rightarrow x = 32.5 / 1.25 = 26$$

Result: The missing base-year quantity for Item C is 26 units.

Problem 6.2: Real Wage / Purchasing Power Deflation

The table below tracks the nominal monthly wages of an industrial worker alongside the Consumer Price Index over a five-year period. Compute the worker's Real Wages and analyze the trend in their actual purchasing power.

Year	Nominal Wage (\$)	CPI (Base 2020=100)
2021	4000	110
2022	4500	125
2023	5000	140
2024	5200	160
2025	6000	175

Solution:

Real wages are calculated using the following deflation formula:

$$\text{Real Wage} = (\text{Nominal Wage} / \text{CPI}) \times 100$$

Year	Nominal Wage	CPI	Real Wage Formula	Real Wage (\$)
2021	4000	110	$(4000/110) \times 100$	3636.36
2022	4500	125	$(4500/125) \times 100$	3600.00
2023	5000	140	$(5000/140) \times 100$	3571.43
2024	5200	160	$(5200/160) \times 100$	3250.00
2025	6000	175	$(6000/175) \times 100$	3428.57

Analytical Insight: Although the worker's nominal salary rose steadily from \$4,000 to \$6,000, inflation outpaced wage growth, leading to a net decline in real purchasing power from \$3,636.36 to \$3,428.57 by 2025.

Problem 6.3: Multi-Item Splicing and Base Shifting

In 2024, a national statistics agency shifts the base year of its price index from 2015 to 2020. Given the old index values below, calculate the spliced new index values under the shifted base.

Year	Old Index (Base 2015=100)
2018	120
2019	135
2020	150
2021	165
2022	180

Solution:

To shift the base to 2020, each index value is divided by the 2020 index value under the old base (150) and multiplied by 100:

$$\text{New Index} = (\text{Old Index Value} / \text{Old Index Value of New Base Year}) \times 100$$

Year	Old Index	Calculation	New Shifted Index (Base 2020=100)
2018	120	$(120/150) \times 100$	80.0
2019	135	$(135/150) \times 100$	90.0
2020	150	$(150/150) \times 100$	100.0
2021	165	$(165/150) \times 100$	110.0
2022	180	$(180/150) \times 100$	120.0

Problem 6.4: Value Index Construction

Calculate the Value Index for the current year based on the data below, tracking changes in total monetary value.

Item	P_0 (\$)	Q_0	P_1 (\$)	Q_1
M	10	5	12	6
N	15	4	20	5

Solution:

The Value Index formula is defined as:

$$V_{01} = (\Sigma P_1 Q_1 / \Sigma P_0 Q_0) \times 100$$

Calculating the components:

$$\Sigma P_1 Q_1 = (12 \times 6) + (20 \times 5) = 72 + 100 = 172$$

$$\Sigma P_0 Q_0 = (10 \times 5) + (15 \times 4) = 50 + 60 = 110$$

$$V_{01} = (172 / 110) \times 100 = 156.36$$